THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

(KATAVI, MBEYA, NJOMBE, RUKWA, RUVUMA AND SONGWE)

FORM FOUR ZONAL MOCK II EXAMINATION

031/1 PHYSICS 1

wazaelimu.com

TIME: 3:00 HOURS **YEAR: 2025**

INSTRUCTIONS

1. This paper consists of section A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in section A and B and **two (2)** question from section C.
3. Section A carries **sixteen (16)** marks, section B carries **fifty-four (54)** marks and section C carries **thirty (30)** marks.
4. All writings must be in **blue** or **black** ink, **except** drawing which must be in pencil.
5. All communication devices and any unauthorized materials are **not** allowed in the examination room.
6. Write your **Examination Number** on every page of your answer sheets provided.
7. Where necessary, the following constants may be used
   (a) Acceleration due to gravity, $g=10m/s^2$
   (b) Speed of sound in air, V=340m/s
   (c) Pie, $\pi = 3.14$
   (d) Speed of light in vacuum = 3×10^8 m/s

SECTION A (16 Marks)

Answer **all** questions in this section.

1. For each of the items (i) – (x) choose the correct answer from the given alternative and write its letter besides the item number in the answer booklet provided.

(i) A rectangular wooden block has a density of 0.8g/cm^3 and dimensions of 0.5m x 0.8m x 0.6m. The maximum pressure it can exert on the ground is ---------- N/m^2.

A 6400 B 48000 C 4000
D 19200 E 400

(ii) The anomalous expansion of water means;

A The maximum temperature of water is 4^0C
B The maximum density of water is 4g/cm^3
C Formation of ice is at 4^0C
D The maximum density of water is at 4^0C
E The maximum density of water is 4g/m^3

(iii) Work is said to be done when:

A Force acts upon a body, but the body does not move
B Force acts upon the body, but the body moves perpendicular to the applied force
C Force acts upon a body and it moves in a direction of applied force
D Force acts upon a body and a body and the body does not move in the direction of applied force
E Force acts upon a body and the body moves.

(iv) What are the two factors that determine buoyancy?

A Volume of liquid displaced and mass of the body
B Weight and mass of the body
C Density of liquid (fluid) and weight of a body
D Volume of a fluid displaced and the density of the fluid
E Mass of the body and density of the body

(v) What is 20^0C in kelvin

A 212 B 70 C 253
D 293 E 100

(vi) The unidirectional property of a p-n junction is useful for its use as a

A Amplifier B Transistor C Transformer
D Oscillator E Rectifier

(vii) Oil is used as a lubricant in a machine because it has

A Low density B High viscosity C Low pressure
D High pressure E Low viscosity

(viii) While Robert is playing with his glitter in a jar of water, he observes the glitter particles moving randomly. He recalls learning about Brownian motion in his Physics class. What does Brownian motion provide evidence of the existence of?

A Friction B Kinetic energy C Atoms
D Gravity E Vector

(ix) Which one is not a region of electromagnetic spectrum?
A Radio waves B X-rays C Infrared light
D Invisible light wave E Ultra violet light

(x) The loudness of a note produced by a vibrating object depends on;
A Number of vibrations per second B The overtones present
C The quality of sound D The wavelength between nodes
E The amplitude of vibrations

2. Match the types of diodes from **List A** with its corresponding function from **List B** by writing the correct letter to the answer sheet provided.

LIST A	**LIST B**
(i) Light emitting diode (ii) Laser diode (iii) Zener diode (iv) Photo diode (v) Semiconductor diode (vi) Avalanche diode	A Fast switching and microwave application B Emits light when an electric current is applied in forward biased C Voltage regulators in different electrical circuits D Used in optical communication as light amplification by stimulating of radiation E Safety electronics like fires and smoke detectors F High frequency oscillators and amplifiers G Provide small voltage to work and very fast switch action H Lower doping concentration compared to Zener diode

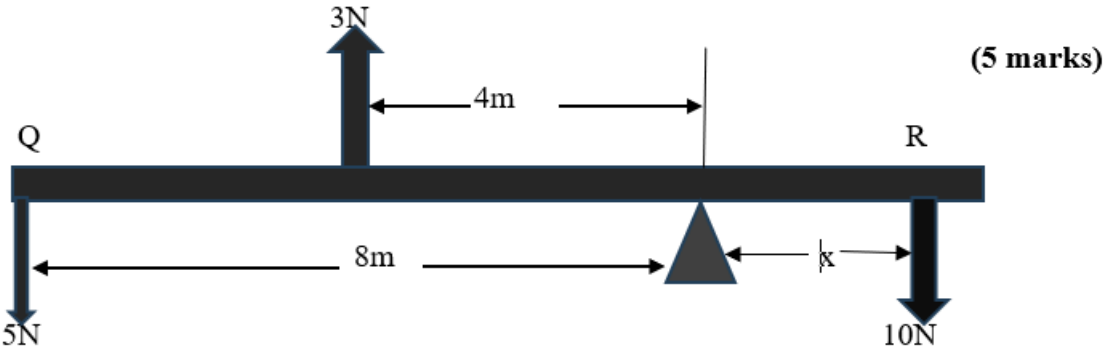
SECTION B (54 Marks)
Answer **all** questions in this section

3. (a) In a light experiment a narrow beam of light directed onto a glass prism leaves the prism and falls on a white screen. Draw a well labelled diagram to show the experimental set – up and observation seen on a screen **(3 marks)**

(b) A lantern projector using a slide of 2cm × 2cm produces a picture 1m × 1m on the screen 12 m from the projection lens. Find the approximate focal length of the projection lens **(6 marks)**

4. (a) A hydraulic press has a large circular piston of radius 0.8m and a circular plunger of radius 0.1m.a force of 200N exerted by the plunger and moved through a distance of 0.64m. find the force exerted on the piston (**4 marks)**

(b) A figure 1 shows the apparatus used by form four students to perform an experiment of investigating the relationship between the moment of force when a beam is at equilibrium, calculate the distance x



(5 marks)

Figure 1

5. (a) While form four students travelling back to school using a car after completion holyday we advised to fasten seat belts. Explain using newton's first law why they have been advised to do so? **(4 marks)**

(b) A block and tackle system of 5 pulleys is used to lift a load of 500N steadily through a height of 20m. the work done against friction is then 2000J. calculate the effort applied. **(5 marks)**

6. (a) Briefly explain two theories used to describe the structure of the solar system. What is their significance? **(4 marks)**

(b) (i) What is global warming? **(1 marks)**

(ii) Name four gases that contribute to global warming. For each gas, write at least one source for producing it. **(4 marks)**

7. (a) (i) How can photographically film, Bubble chamber and gold leaf electroscope as devices be used to detect radioactive? **(3 marks)**

(ii) A sample contains 800g of iodine 131. How much the sample will remain undecayed after 40 days? The half-life of iodine 131 is 8 days. **(4 marks)**

(b) What are the major two applications of nuclear fission in daily life? **(2 marks)**

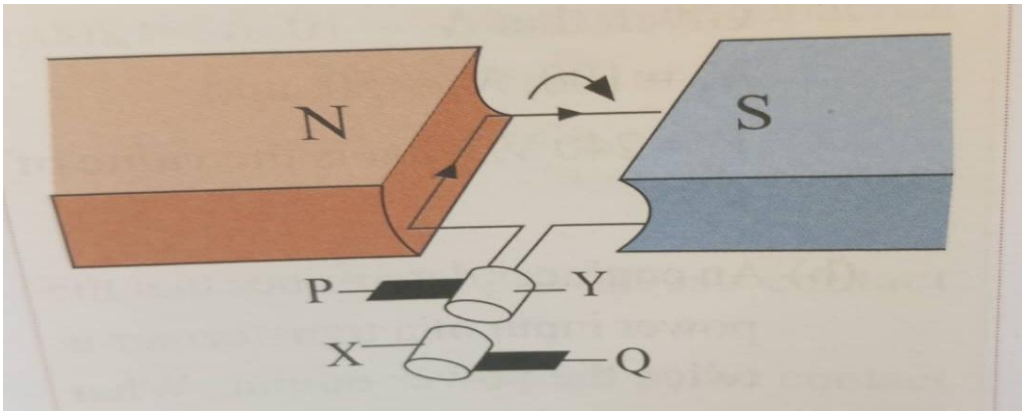
8. (a) Explain four factors that affect the resistance of conductor. **(4 marks)**

(b) The parallel combination of 2Ω and 5Ω resistors are connected across the external resistor of 3Ω through the supply of 4V. draw the diagram and find the current flowing through a 5Ω resistor. **(5 marks)**

SECTION C (30 Marks)

Answer only **two (2)** questions from this section

9. (a) The figure below shows a coil in a magnetic field. The coil is rotated in the direction shown by the arrow.



(i) Name the parts labelled X and Y **(1 mark)**

(ii) State two factors that affect the magnitude of the e.m.f produced by rotating the coil in the figure above **(3 marks)**

(iii) Explain the changes that should be made in the set – up for it to produce a direct current **(3 marks)**

(b) A 20-watt lamp with a resistance of 7.2Ω uses a power supply from the secondary coil of the transformer. If the primary coil is connected to a 120V source,

(i) What is the ratio of number of turns on the primary coil to the number of turns on the secondary coil **(5 marks)**

(ii) Is the transformer step up or step down Give a reason to your answer **(3 marks)**

10. (a) Distinguish the concepts of conducts, semiconductors and insulators in terms of Their energy bands. **(6 marks)**

(b) Describe the mechanism of doping intrinsic semiconductor to obtain, p-type semiconductor. **(6 marks)**

(c) Why most of the transistor in use are n-p-n transistors? **(3 marks)**

11. (a) The shortest length of the air column in a resonance tube (with one end closed and one end opened) which resonate to a note of frequency 500Hz is found to be 160mm.

(i) What is the wavelength of this sound in air? **(3 marks)**

(ii) Under the same condition what is the shortest length of the column if the note frequency is 800Hz? **(3 marks)**

(b) Three capacitors of capacitance A, B and C are arranged to have minimum value. their capacitance is given as 10μF,20μF and 30μF respectively. Calculate the value of a single capacitor that replace them. **(5 marks)**

(c) Explain four ways on how to take care for the lead acid batteries to extend their life span **(4 marks)**